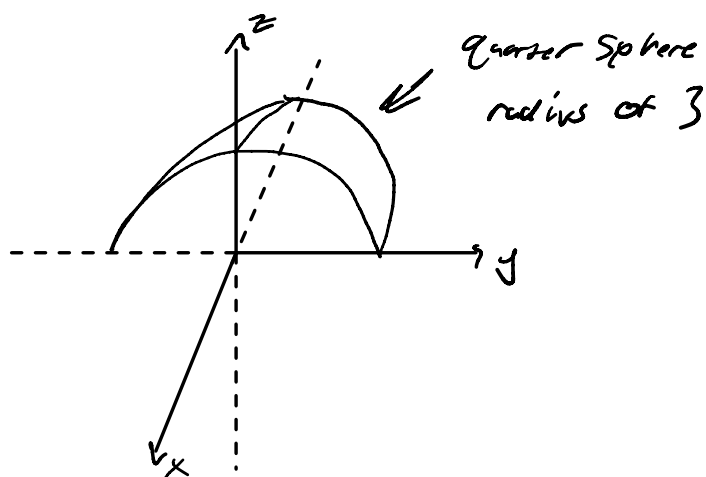


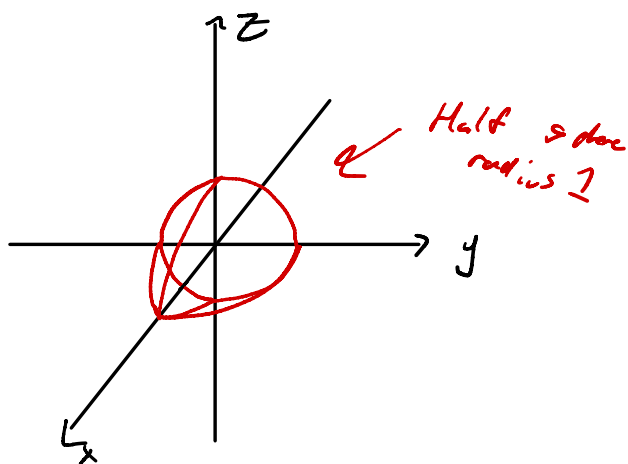
15.8 Triple Integrals in Spherical Coordinates

Prerequisite information for this section: For each integral below, **sketch** the (three-dimensional) region of integration. Be sure to note the order of integration before beginning each sketch.

$$\int_0^3 \int_{-\sqrt{9-x^2}}^0 \int_0^{\sqrt{9-x^2-y^2}} f(x, y, z) dz dy dx$$



$$\int_{-1}^1 \int_0^{\sqrt{1-y^2}} \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} f(x, y, z) dz dx dy$$



Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

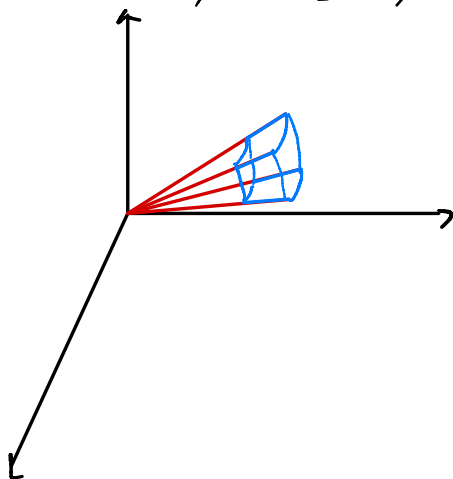
- **WARNING:** [Give the warning from example 2 of Section 15.8 of your text.]

There is not universal agreement on notation for spherical coords -
most physics books reverse θ and ϕ and use r instead of ρ .

- **Spherical wedge:** [Give the set notation and sketch for this item, Figure 7 and the paragraph above this figure.]

$$E = \{(\rho, \theta, \phi) \mid \rho \in [a, b] \wedge \theta \in [\alpha, \beta] \wedge \phi \in [c, d]\}$$

where $a \geq 0$, $\beta - \alpha \leq 2\pi$, and $d - c \leq \pi$



Topics to learn in this section:

1. The geometry of the spherical coordinate system.
2. The basic shapes of solids in spherical coordinates.
3. The idea that the spherical coordinate system can be used to simplify equations and volume integrals of certain three-dimensional surfaces and solids.

15.9 Change of Variables in Multiple Integrals

$$\frac{35}{73}$$

Prerequisite information for this section: Calculate the following cross product and determinants.

$$1. \langle 4, -5, -6 \rangle \times \langle 7, 2, 0 \rangle = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -5 & -6 \\ 7 & 2 & 0 \end{vmatrix} = 12\hat{i} - 42\hat{j} + 43\hat{k}$$

$$2. \begin{vmatrix} 1 & 1 & 1 \\ 4 & -5 & -6 \\ 7 & 2 & 0 \end{vmatrix} = 12 - 42 + 43 = 13$$

$$3. \begin{vmatrix} 2 & 3 & -4 \\ x & y & z \\ -1 & -2 & 5x \end{vmatrix} = 2 \begin{vmatrix} y & z \\ -2 & 5x \end{vmatrix} - 3 \begin{vmatrix} x & z \\ -1 & 5x \end{vmatrix} - 4 \begin{vmatrix} x & y \\ -1 & -2 \end{vmatrix}$$

$$= 2(5xy + 2z) - 3(5x^2 + z) - 4(-2x + y)$$

Definitions in this section are going to be discussed as we get to them in lecture.